

Getting Start

Before you begin, ensure you have the following:

- Plant Watering Kit with Arduino UNO R4 WiFi
- A computer with internet access
- USB-C programming cable for Arduino
- Basic knowledge of Arduino programming

Arduino IDE Setup

- Download and install the latest version of the Arduino IDE from the official Arduino website, or install the prepared installation package in the USB flash drive coming with the package.

Preparation

Download Arduino IDE Software

- Download Arduino Ide installation package from [official web site](#) Or find the installation package in the microSD card which is coming with the kit.

If you want to use the latest package, please download it from official web site. and select the OS platform that you are using, for example: windows, Linux or Mac OS.

Before using this kit, you need to know about the main board of this kit.

Know about Arduino UNO R4 WiFi board

The Arduino UNO R4 Wi-Fi is designed around the 32-bit microcontroller RA4M1 from Renesas, which also features an ESP32 module for Wi-Fi® and Bluetooth® connectivity. Its distinctive 12x8 LED matrix allows for prototyping visuals directly on the board, and with a Qwiic connector, you can create projects in a plug-and-play style.

Features

- The MCU on the board is the high-performance Renesas RA4M1 (Arm® Cortex®-M4) with a 48 MHz clock speed, 32 kB SRAM, and 256 kB flash memory.
- This MCU features an RTC, a DAC, a CAN bus, and has support for HID via USB.

- The UNO R4 Wi-Fi also features an ESP32-S3 for Wi-Fi®/Bluetooth® connectivity, which can also be separately programmed via a specific header.



Arduino Cloud

The UNO R4 WiFi is compatible with the Arduino Cloud platform. Build IoT projects in just minutes!

[GO TO PLATFORM](#)



Cheat Sheet

A reference to all technical features on this board, with pointers to additional documents.

[CHEAT SHEET](#)



5 V Operating Voltage

The RA4M1 and the GPIOs of this board operates on 5 V.



LED Matrix

Learn how to create animations and graphics on the 12x8 LED matrix.

[DOCUMENTATION](#)



Real-Time Clock (RTC)

Keep track of time & date and set alarms with the built-in RTC.

[DOCUMENTATION](#)



Digital-to-Analog Converter (DAC)

Use the onboard 12-bit DAC to build sophisticated audio projects.

[DOCUMENTATION](#)



Mouse/Keyboard Emulation (HID)

Build game controllers by emulating a mouse/keyboard.

[DOCUMENTATION](#)



Input Voltage

Power your UNO R4 board through the VIN pin or the barrel jack at up to 6-24 V.

Tech Specs

Here are the technical specifications for the Arduino® UNO R4 Wi-Fi.

Notes

- **ESP header:** The ESP32-S3 module on this board operates on 3.3 V. The ESP header located close to the USB-C® connector is 3.3V only and should not be connected to 5 V. Connecting to 5V may damage your board.
- **Qwiic connector:** The Qwiic connector on this board is connected to a secondary I2C bus, I2C0. This connector is 3.3 V only; connecting higher voltages may damage your board. To initialize this bus, use `Wire1.begin()` instead.
- **Maximum current draws per pin:** The UNO R4 series' maximum current draw per GPIO is 8 mA, which is significantly lower than previous versions. Exceeding this limit may damage your pin/board.

Microcontroller	
Renesas RA4M1 (Arm® Cortex®-M4)	
Radio Module	
ESP32-S3-MINI-1-N8	
USB	
USB-C®	Programming Port

USB	
USB-C®	Programming Port
Pins	
Digital I/O Pins	14
Analog input pins	6
DAC	1
PWM pins	6
External interrupts	2,3

Communication	
UART	Yes, 1x
I2C	Yes, 1x
SPI	Yes, 1x
CAN	Yes 1 CAN Bus

Power	
Circuit operating voltage	5 V (ESP32-S3 is 3.3 V)
Input voltage (VIN)	6-24 V
DC Current per I/O Pin	8 mA

Clock speed	
RA4M1	48 MHz
ESP32-S3-MINI-1-N8	up to 240 MHz
Memory	
RA4M1	256 kB Flash, 32 kB RAM
ESP32-S3-MINI-1-N8	384 kB ROM, 512 kB SRAM

Software

The following software tools allow you to program your board both online and offline.



Arduino IDE



Arduino CLI



Web Editor

Suggested Libraries

- **Wire:** This library allows you to communicate with I2C devices, a feature present on all Arduino boards. I2C is a very common protocol, primarily used for reading/sending data to/from external I2C components. [More info](#)
- **SPI:** This library allows you to communicate with SPI devices, with the Arduino as the controller device. This library is bundled with every Arduino platform (avr, megaavr, mbed, samd, sam, arc32), so you do not need to install it separately.
- **Servo:** Allows Arduino boards to control a variety of servo motors. This library can control a great number of servos, making careful use of timers: it can control 12 servos using only 1 timer. On the Arduino Due, you can control up to 60 servos.

Download Arduino IDE

- Official website: arduino.cc/en/software

Downloads



Arduino IDE 2.3.2

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

Nightly builds with the latest bugfixes are available through the section below.

SOURCE CODE

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

DOWNLOAD OPTIONS

Windows Win 10 and newer, 64 bits
Windows MSI installer
Windows ZIP file

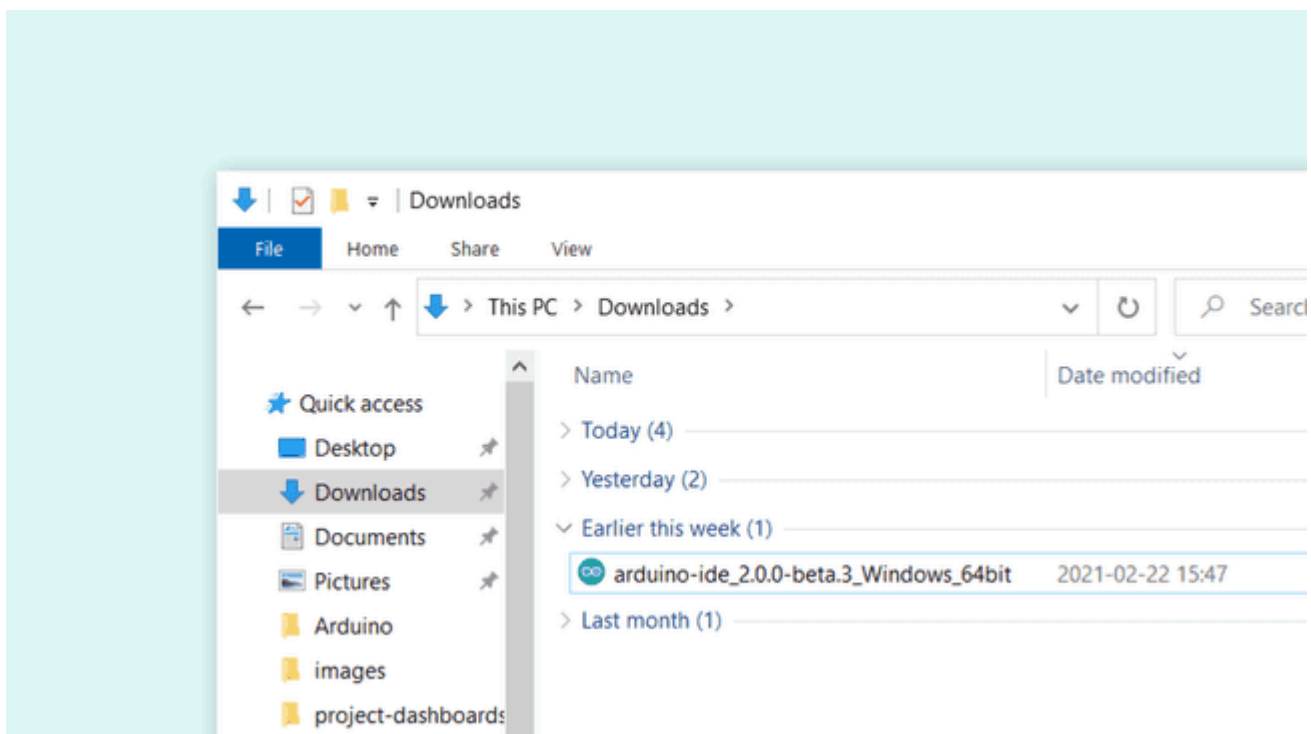
Linux AppImage 64 bits (X86-64)
Linux ZIP file 64 bits (X86-64)

macOS Intel, 10.15: "Catalina" or newer, 64 bits
macOS Apple Silicon, 11: "Big Sur" or newer, 64 bits

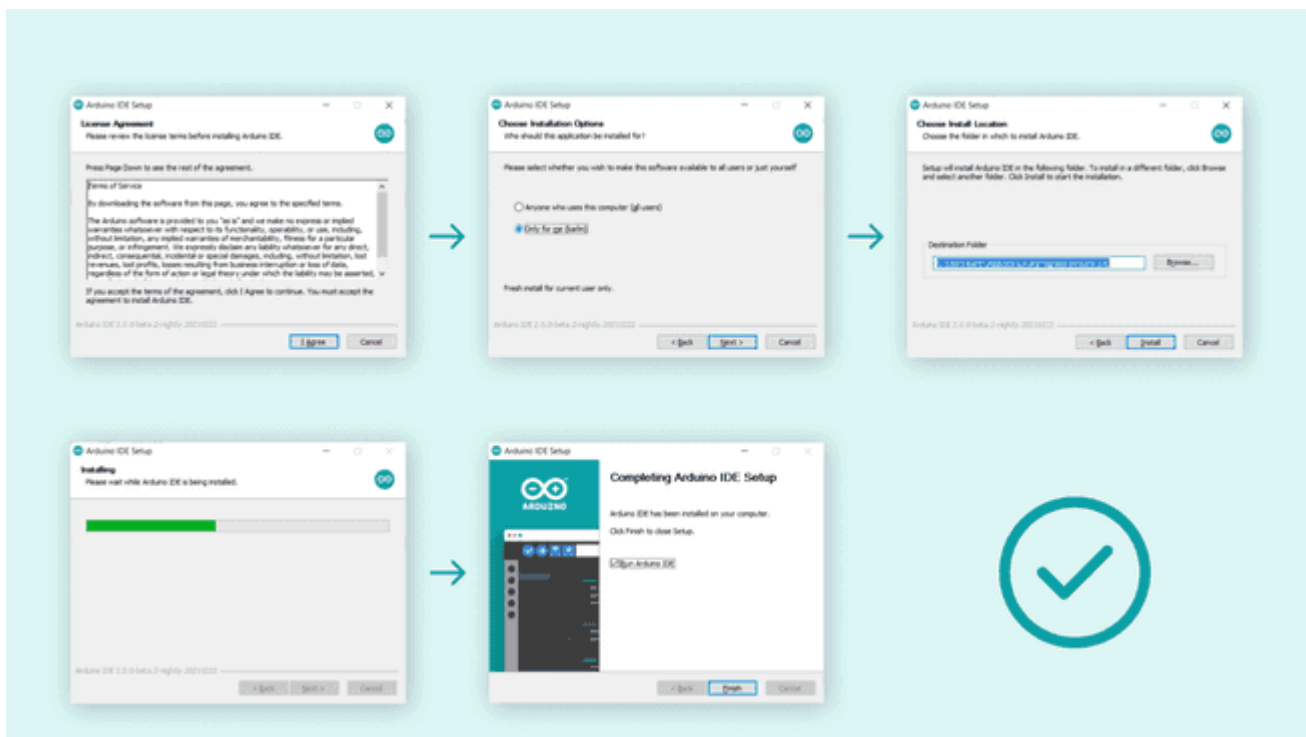
[Release Notes](#)

Installing the Arduino IDE 2

- Please refer to this URL: [Downloading and installing the arduino IDE 2](#)
- Windows To install the Arduino IDE 2 on a Windows computer, simply run the file downloaded from the software page



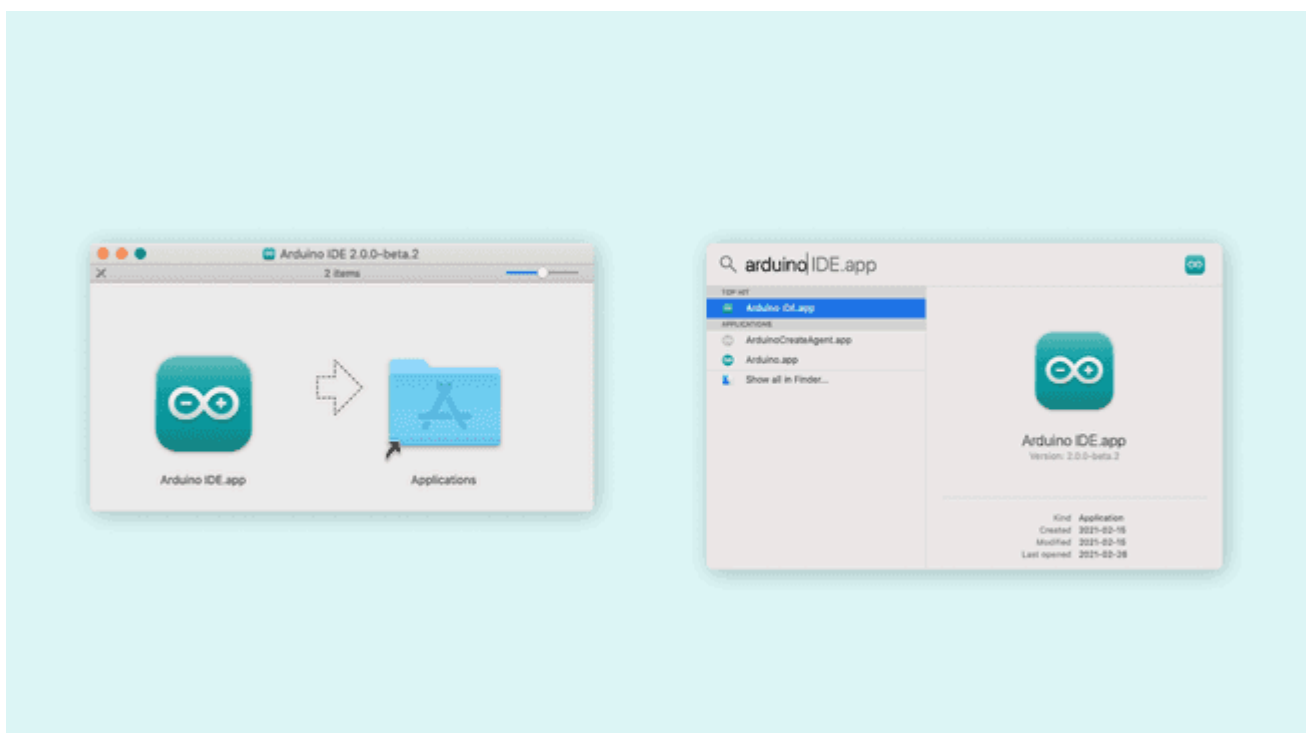
Follow the instructions in the installation guide. The installation may take several minutes.



You can now use the Arduino IDE 2 on your Windows computer!

- MacOS

To install the Arduino IDE 2 on a macOS computer, simply copy the downloaded file into your application folder.

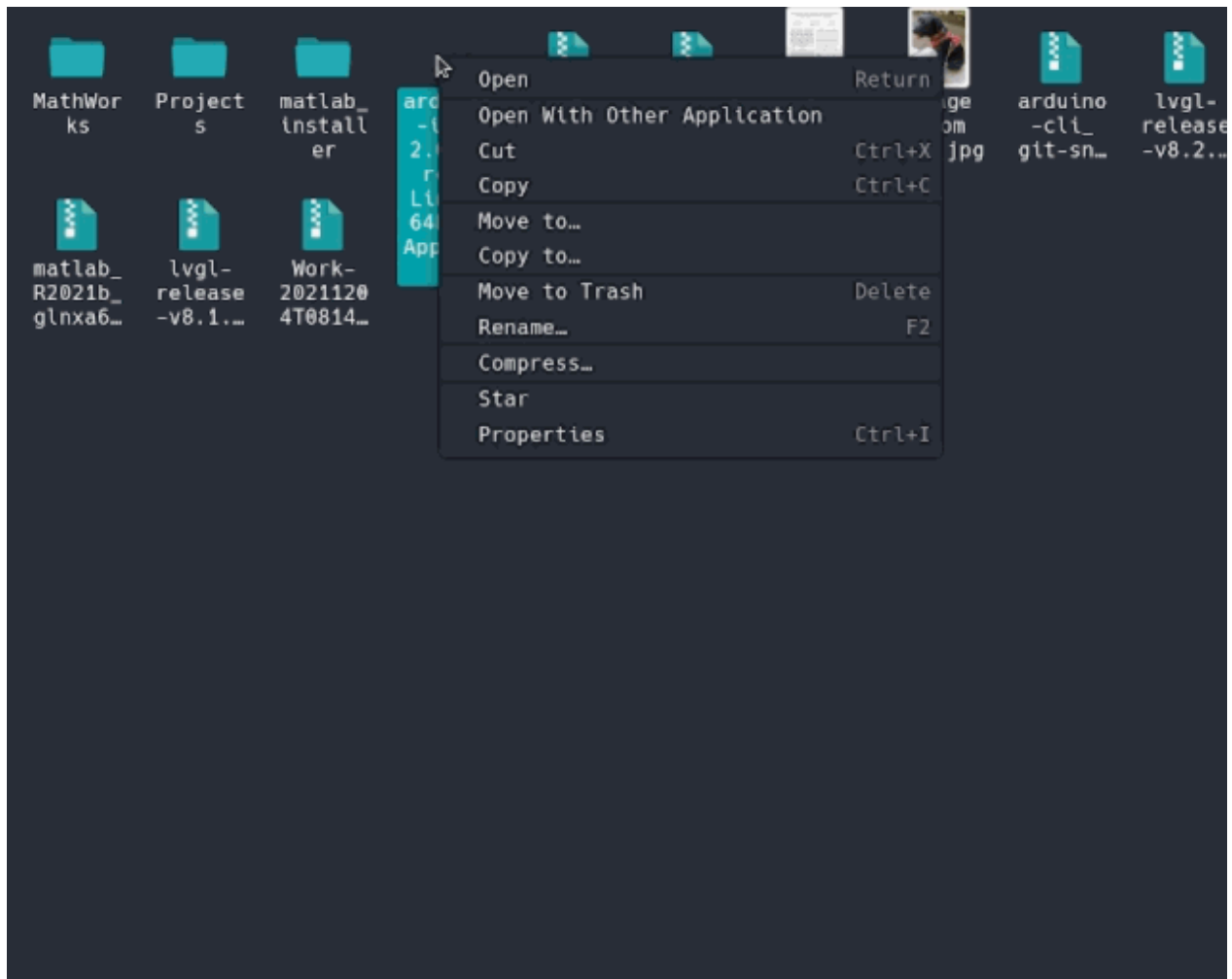


You can now use the Arduino IDE 2 on your macOS computer!

- Linux To install the Arduino IDE 2 on Linux, first download the ApplImage 64 bits (X86-64) from the [Arduino Software](#) page.

Before we can launch the editor, we need to first make it an executable file. This is done by:

1. `right-click` the file,
2. `choose Properties`,
3. `select Permissions tab`,
4. `tick the Allow executing file as program box`.



You can now double click the file to launch the Arduino IDE 2 on your Linux machine. In case you cannot run the AppImage file, make sure that FUSE is installed on your system.

In Ubuntu (>= 22.04):

```
sudo add-apt-repository universe
sudo apt install libfuse2
```

In Fedora:

```
dnf install FUSE
```

See instructions for installing FUSE on your distribution [here](#)

To enable the Arduino IDE to access the serial port and upload code to your board, the following rule can be added to `/etc/udev/rules.d/99-arduino.rules`.

```
SUBSYSTEMS=="usb", ATTRS{idVendor}=="2341", GROUP="plugdev", MODE="0666"
```

How to upload a sketch with the Arduino IDE 2

In the Arduino environment, we write sketches that can be uploaded to Arduino boards. In this tutorial, we will go through how to select a board connected to your computer, and how to upload a sketch to that board, using the Arduino IDE 2. You can easily download the editor from the [Arduino Software](#) page. You can also follow the [downloading and installing the Arduino IDE 2 tutorial](#) for more detailed guide on how to install the editor.

Requirements

- Arduino IDE 2 installed.

Verify VS Upload

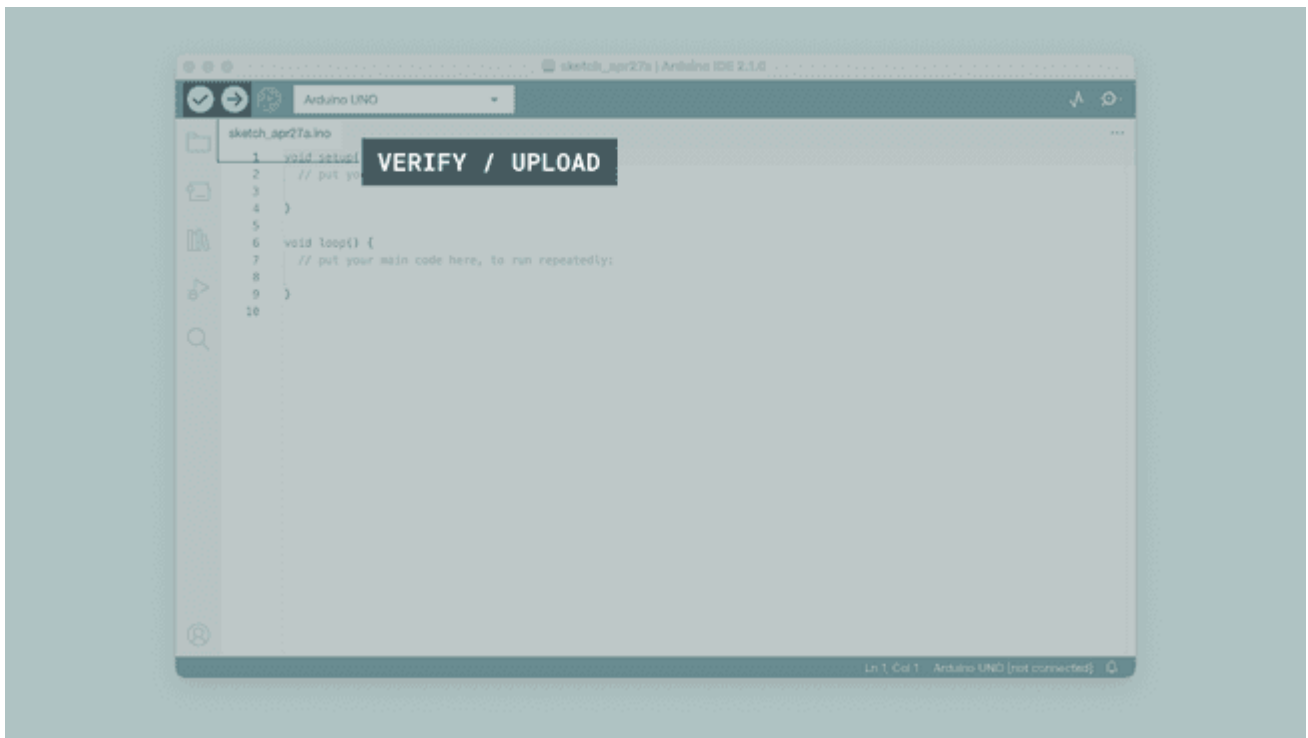
There are two main tools when uploading a sketch to a board: `verify` and `upload`. The verify tool simply goes through your sketch, checks for errors and compiles it. The upload tool does the same, but when it finishes compiling the code, it also uploads it to the board.

A good practice is to use the verifying tool before attempting to upload anything. This is a quick way of spotting any errors in your code, so you can fix them before actually uploading the code.

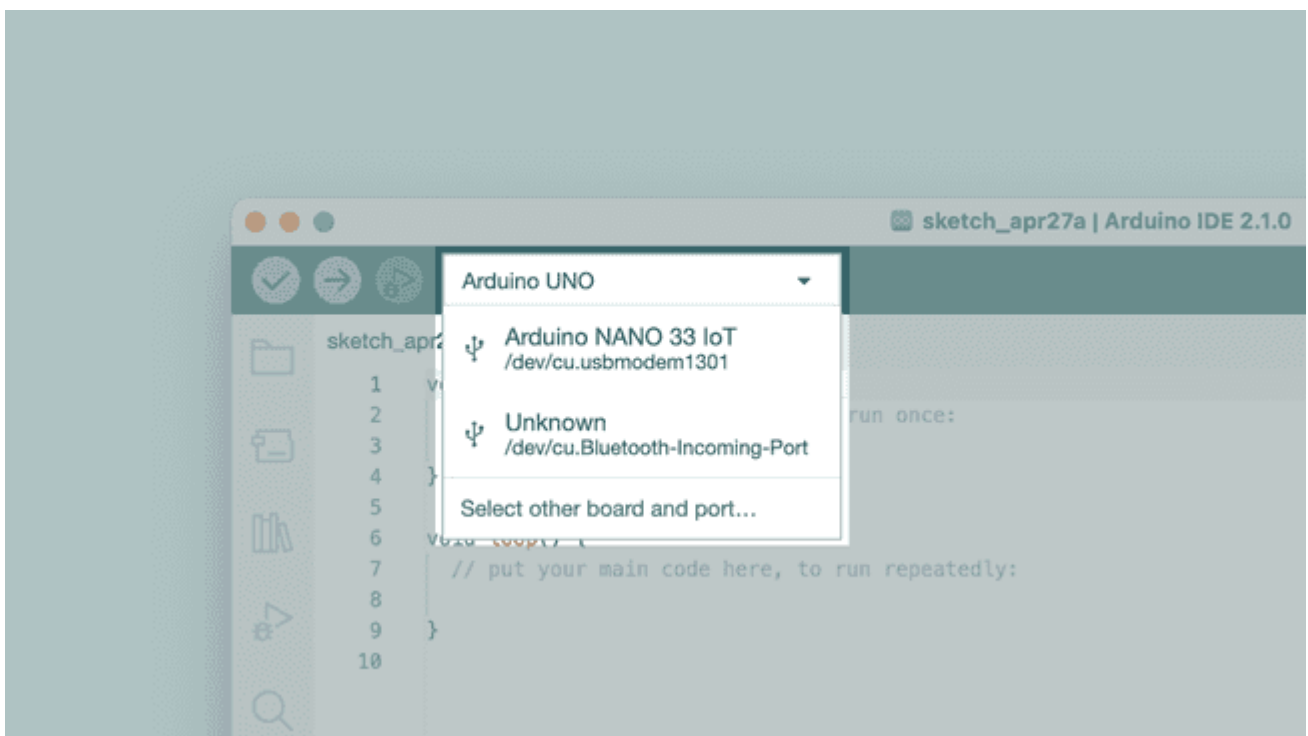
Uploading a Sketch

Uploading a sketch is quick and easy, but let's take a look at what we need to do.

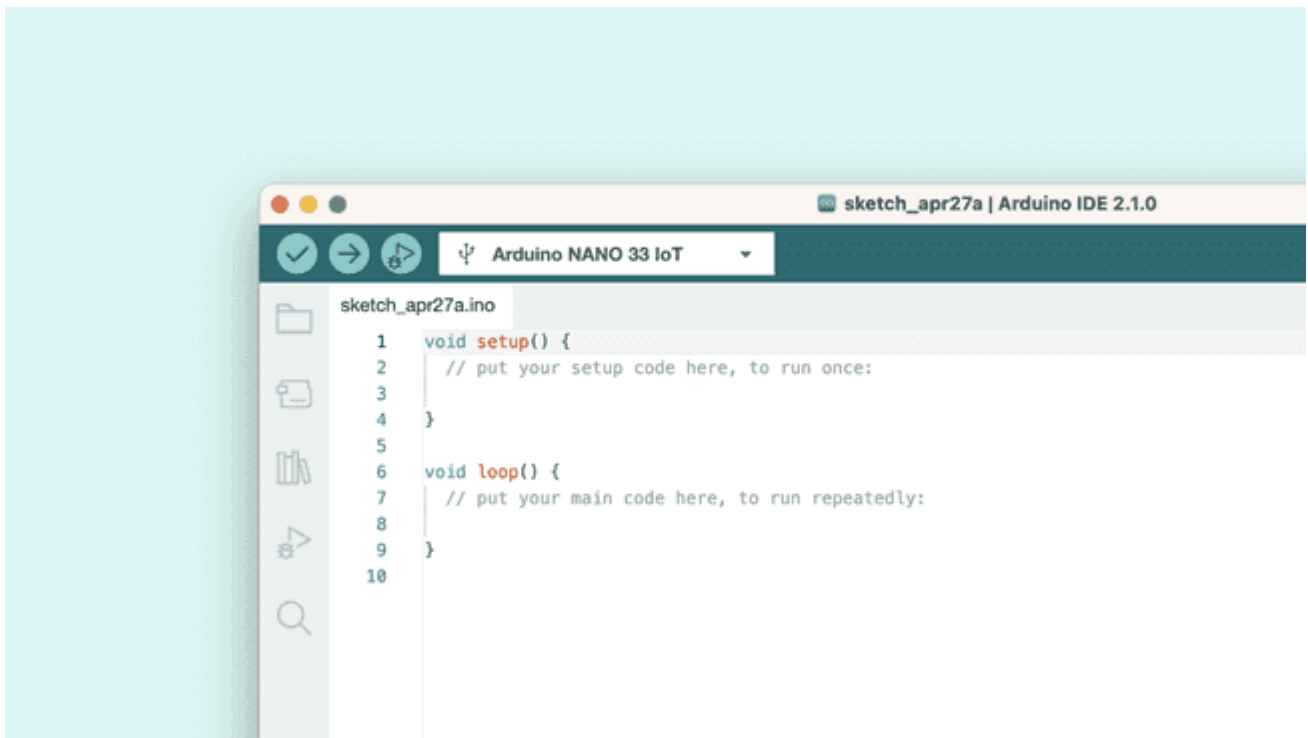
- Open the Arduino IDE 2.
- With the editor open, let's take a look at the toolbar at the top. At the very left, there is a `checkmark` and an `arrow pointing right`. The checkmark is used to `verify`, and the arrow is used to `upload`.



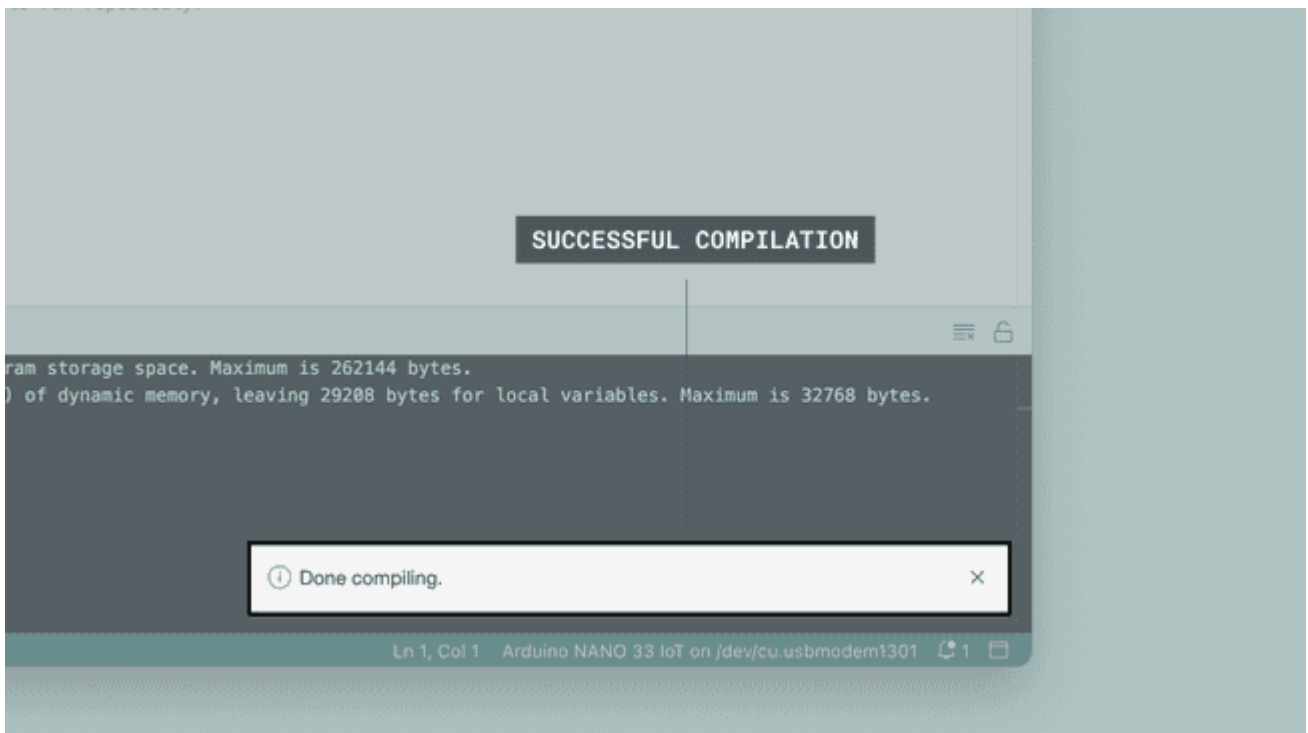
- But before doing any of that, we should choose what board we are uploading to. Beside the verify and upload button you should see a drop down menu, this will, in most cases, display Arduino boards that are connected to your computer. If your board is not automatically detected, you can either press "Select other board and port..." in the drop-down and follow the instructions, or go to **Tools** > **Board and Tools** > **Port** in the toolbar menu to select the board and port manually.



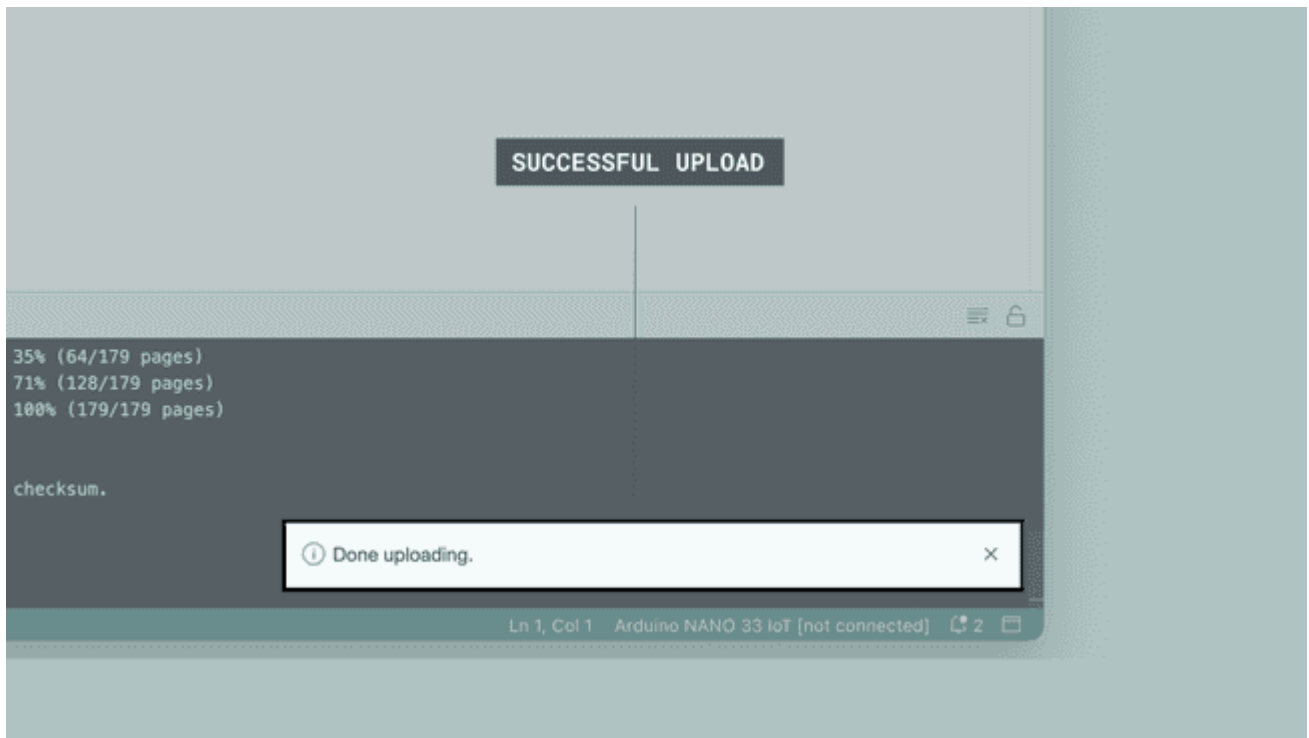
So connect your board to the computer and select it from the drop down menu. You'll know that there is a connection to the board when the board name appears in **bold**.



- Click on the verify tool (checkmark). Since we are verifying an empty sketch, we can be sure it is going to compile. After a few seconds, we can see the result of the action in the console (black box in the bottom).



- With the board selected, we are good to go! Click on the upload button, and it will start uploading the sketch to the board.
- When it is finished, a notification pops up in the bottom right of your IDE window. Of course, sometimes there are some complications when uploading, and these errors will be listed here as well.



Congratulations, you have now uploaded a sketch to your Arduino board!

**** Please Read (Only Linux Users) ****

It might happen that when you upload a sketch - after you have selected your board and the serial port - you get an error such as "... Permission denied". If you get this error, you need to set serial port permission.

To do this, open your Terminal and type:

```
ls -l /dev/ttyACM*
```

you will get something like:

```
crw-rw---- 1 root dialout 188, 0 5 apr 23.01 ttyACM0
```

The "0" at the end of ACM might be a different number, or multiple entries might be returned. The data we need is "dialout" (is the group owner of the file).

Now we just need to add our user to the group:

```
sudo usermod -a -G dialout <username>
```

where is your Linux user name. ** You will need to log out and log in again for this change to take effect ** .

```
osboxes@osboxes: ~  
osboxes@osboxes:~$ ls -l /dev/ttyACM*  
crw-rw---- 1 root dialout 166, 0 Feb 23 15:05 /dev/ttyACM0  
osboxes@osboxes:~$ sudo usermod -a -G dialout osboxes  
[sudo] password for osboxes:  
osboxes@osboxes:~$
```

After this procedure, you should be able to proceed normally and upload the sketch to your board or use the Serial Monitor.